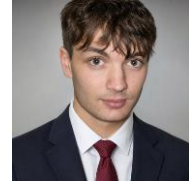


Marcos Flores Panadero

Mechanical Engineering · ETSIDI – UPM · Madrid, Spain
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Profile

Third-year Mechanical Engineering student (ETSIDI – UPM) with hands-on experience in CAD design, CNC machining, industrial automation and materials analysis. Proven ability to work in technically demanding environments and to communicate complex concepts clearly. Interested in aerospace, defence and high-tech industrial sectors. English B2 certified. Available from 7th July 2026.

Education

B.Eng. in Mechanical Engineering

2022 – present

Escuela Técnica Superior de Ingeniería y Diseño Industrial (ETSIDI) · UPM

- 3rd year. Key subjects: Machine Design, Theory of Machines and Mechanisms, Manufacturing Technology, Engineering Graphics, Strength of Materials, Fluid Mechanics, Thermodynamics, Materials Science, Automation & Control.
- English B2 — LanguageCert.

Experience

Private tutor — Excel and Mathematics

2022 – present

Self-employed · Madrid

- One-to-one and group lessons in advanced Excel and Mathematics for secondary and pre-university students.
- Strong ability to explain complex technical concepts clearly and adapt to each student's level.

Technical projects

CNC machining of a mechanical part — full workflow

ETSIDI – UPM

- 3D part design in Autodesk Inventor and technical drawings in AutoCAD per ISO standards.
- Tool selection with Sandvik Coromant and CAM toolpath programming in Fusion 360.
- Real CNC machining on a lathe/milling machine with WinUnisoft control.

Orange juicer — mechanical design

ETSIDI – UPM

- Full 3D modelling of the assembly in Autodesk Inventor: parts, tolerances, assembly constraints and animated assembly video.
- Workshop drawings with functional and geometric dimensioning per ISO standards.

Industrial automation with Siemens PLC — Step 7

ETSIDI – UPM

- Siemens S7 PLC programming in Step 7: control logic, GRAFCET diagrams, OB1/OB100 blocks and PLCSIM simulation.
- Automated sequential control of double-acting pneumatic cylinders on a physical test bench.

Materials selection — Ansys Granta

ETSIDI – UPM

- Analysis and selection of materials for mechanical components using Ansys Granta, applying Ashby's methodology based on strength, weight and cost criteria.

Technical skills

CAD Design: Autodesk Inventor, AutoCAD (3D modelling and 2D drawings)

CAM / CNC: Fusion 360 (CAM), WinUnisoft, lathe and milling machine

Automation: Step 7 (Siemens), PLC programming, GRAFCET

Simulation: MATLAB, MITCalc, SAM, Ansys Granta

Technical base: Fluid mechanics, thermodynamics, strength of materials, materials science

Manufacturing: Welding, metrology

IT: Excel (advanced), Word, PowerPoint

Languages: Spanish (native) · English B2 — LanguageCert

A d d i t i o n a l i n f o r m a t i o n

Personal website: marcosflorespanadero.com · **LinkedIn:** linkedin.com/in/marcosflorespanaderoigmec

Available for summer internship — from 7th July 2026.